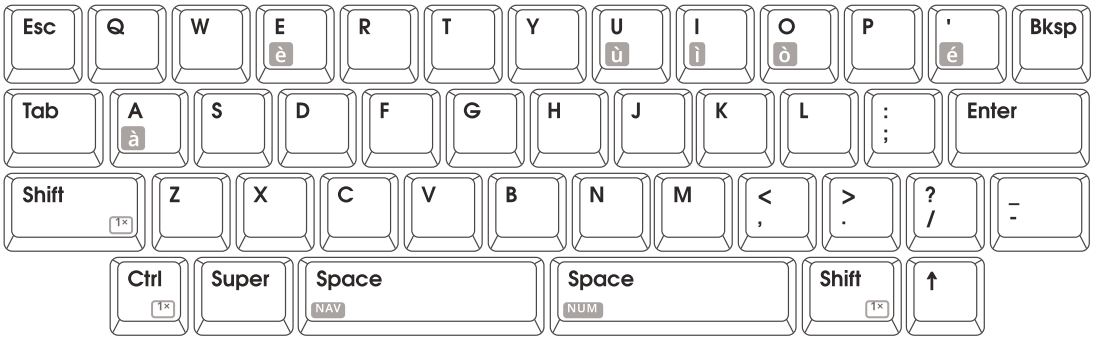
Agar mini

Custom ZMK Layout
Key Function Manual

ZMK FIRMWARE

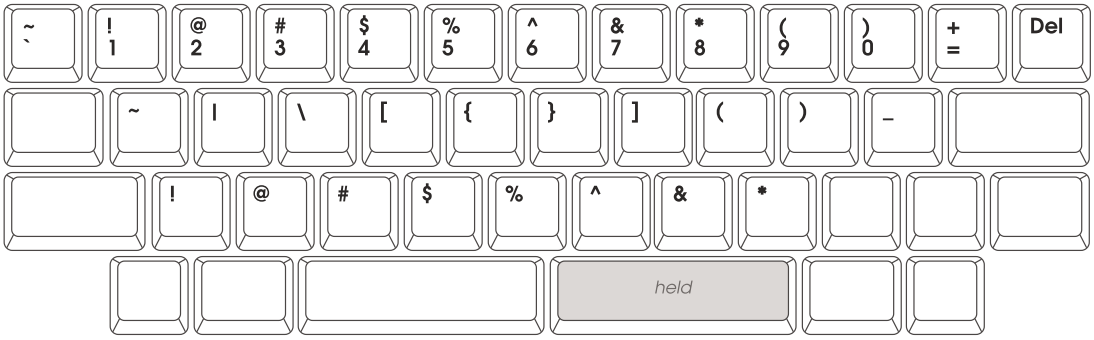
Quick tip: a **grey tag** is what the key does when held. **1×** marks a sticky key: tap it, then the next key. Blank keys pass through to Layer 0.

Layer 0 BASE



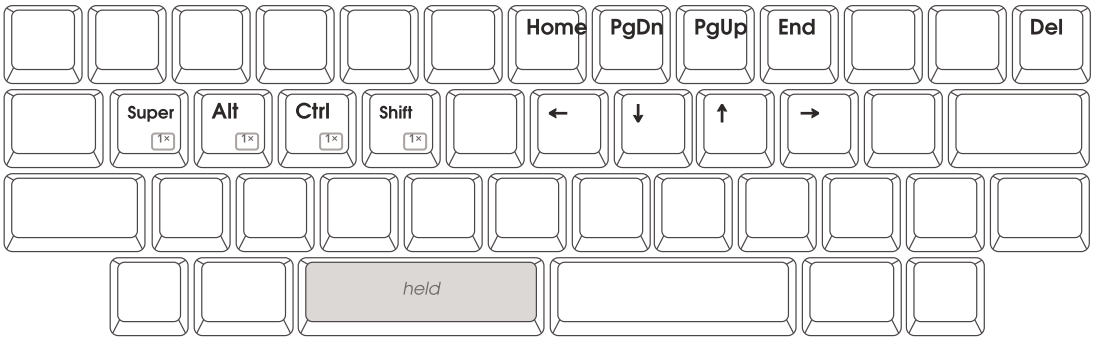
- è — **Hold for the accent** à è ì ò ù, and é on '
- 1× — **Sticky:** tap, then the next key
- NUM — **Hold for Layer 1**, right Space
- NAV — **Hold for Layer 2**, left Space
- Both Spaces — **Both Spaces** for Layer 3
- ' — **Plain apostrophe**; Shift gives "
- Super — **Plain key**, hold it
- ↑ — **Up arrow** for the last command

Layer 1 NUM



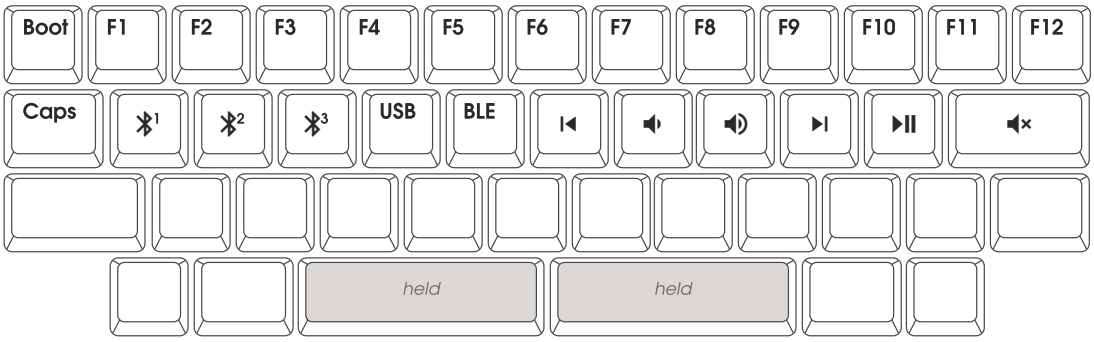
- Hold right Space — **Hold right Space**
- 1–0 — **Super + number** switches mango workspace
- . / - — **Pass through** for ~/, 1.5, IPs
- ` ~ ^ — **Plain characters**, not dead keys

Layer 2 NAV



- Hold left Space — **Hold left Space**
- ← ↓ ↑ → — **Arrows on hjkl**
- 1× — **Sticky modifiers:** tap D, tap L = Ctrl+→
- Home — **Home, PgDn, PgUp, End** sit above the arrows

Layer 3 FN



- Both Spaces — **Hold both Spaces**
- BT1 — **Bluetooth profile 1–3**; hold 3 s to clear its pairing
- USB — **Switch output** to USB or BLE
- Boot — **Bootloader:** hold 3 s; a tap restarts
- ⏮ ⏯ ⏭ — **Media** previous, play/pause, next
- 🔉 🔊 🔇 — **Volume** down, up, mute

Everyday actions

ACTION	SENDS	KEYS
è à ì ò ù		hold the vowel
é		hold [']
È		tap [Shift], hold [E]
Capital letter		tap [Shift] (right thumb), tap the letter
Apostrophe / quote	`' / "`	tap ['] / tap [Shift], tap [']
Newline in Claude Code	`Shift+Enter`	tap [Shift], tap [Enter]
Question / exclamation mark	`? / !`	tap [Shift], tap [/] / hold [R Space] + [Z]
Number	`1 … 0`	hold [R Space] + [Q] … [P]
Backspace		top-right corner, as on the Q4
Run the last command again	`Up, Enter`	tap [↑] (bottom right), tap [Enter]
Copy / paste	`Ctrl+C / Ctrl+V`	tap [Ctrl], tap [C] / [V]
Kitty copy / paste	`Ctrl+Shift+C / V`	tap [Ctrl], tap [Shift], tap [C] / [V]
Switch workspace	`Super+3`	hold [Super] + hold [R Space] + [E]
Move window to workspace	`Super+Shift+3`	tap [Shift], then hold [Super] + hold [R Space] + [E]
Other mango shortcuts	`Super+Return, Super+q`	hold [Super] + the key
Focus direction	`Super+Left`	hold [L Space], tap [A], tap [H]
Kitty tab	`Alt+3`	hold [L Space], tap [S] (Alt), then hold [R Space] + [E]
Accept fish autosuggestion	`Right`	hold [L Space] + [L]

No mango config changes are needed: every action sends exactly what a full-size keyboard would.

Timing

KEY	FLAVOR	TERM	QUICK-TAP	PRIOR IDLE
Space / NAV	balanced	200	175	150
Space / NUM	balanced	200	175	—
Vowels and '	tap-preferred	250	—	—

STICKY KEYS	RELEASE AFTER	QUICK RELEASE
Shift, Ctrl, Alt, Super	1500	yes

- All values in milliseconds. Vowels are decided by time only: shorter than 250 ms types the letter, longer gives the accent; they type on release. With Ctrl, Alt or Super active, a long press gives the plain letter, so shortcuts like Ctrl+A still work.
- Sticky keys apply to the next key only and expire after 1.5 s. Tapping Ctrl then Shift stacks them. They can also be held normally.
- Space / NUM has no prior-idle setting; otherwise typing `port 8080` quickly would give letters instead of digits.

Design

1. **Thumbs hold, nails tap one key at a time.** No pinkies, no ring-finger chords; ring fingers only reach Esc and Backspace.
2. **Long-press vowels** for Italian accents, like a phone keyboard.
3. **Sticky Shift, Ctrl and Alt:** tap, then the next key.
4. **' and " are always plain**, despite `us intl` dead keys.
5. **Super under the left thumb, a second Shift under the right.** Backspace stays top-right.

Requires `us intl`

Accents and plain quotes are typed as `us intl` dead-key sequences: ` + vowel for è, ' + e for é, ' + Space for '. On a device with a plain US layout, accents come out as `e and quotes get an extra space.

Dropped from stock

Mouse keys (trackpad available), Win+V, Print Screen (mango uses Super+Shift+S), Insert, Pause, Scroll Lock, right Alt, arrow cluster (↑ kept bottom right, the rest on NAV).